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Geometry of Linear Equations, Matrices, and Matrix Operations

Geometry of linear equations in 2 and 3 unknowns

Example 1:

Consider the system of equations:

$$l_1: 2x_1 + x_2 = 3$$
$$l_2: x_1 + 2x_2 = 3$$

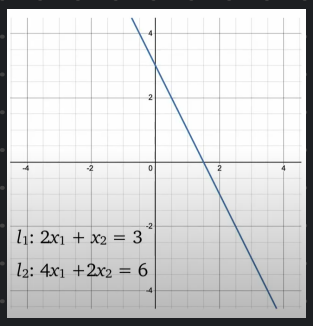
These lines intersect at $(1,1)$. $(x_1, x_2) = (1,1)$ is the soln we want. These have a unique soln.

Example 2:

$$l_1: 2x_1 + x_2 = 3$$
$$l_2: 4x_1 + 2x_2 = 6$$

} These lines overlap.

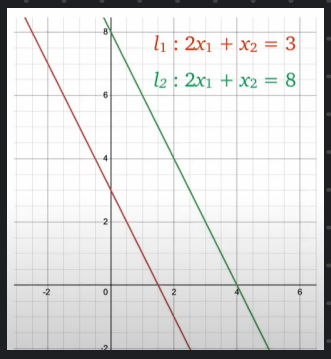
They have ∞ly many solutions.



Ex. 3.

$$2x_1 + x_2 = 3$$
$$2x_1 + x_2 = 8$$

] ← These lines are parallel. They do not have a solution.



Consistent system: A system of linear equations is said to be consistent if it has 1/many solutions.

Inconsistent system: A system of linear equations is said to be consistent if it has no solutions.

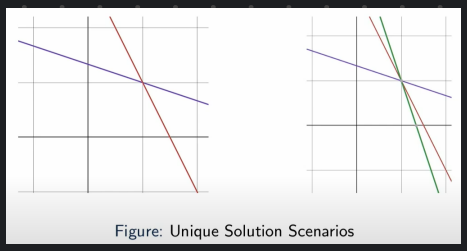


Figure: Unique Solution Scenarios

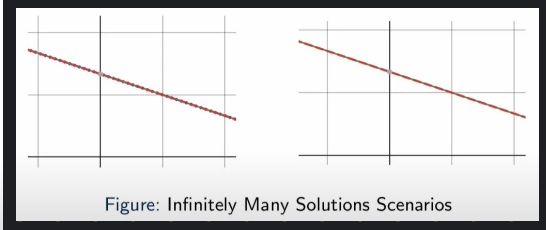


Figure: Infinitely Many Solutions Scenarios

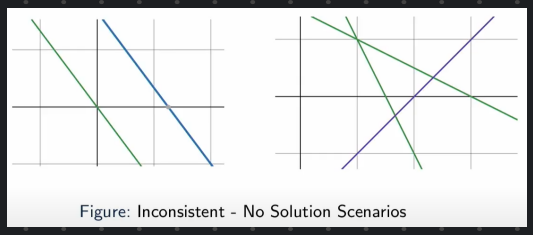


Figure: Inconsistent - No Solution Scenarios

Measurement error

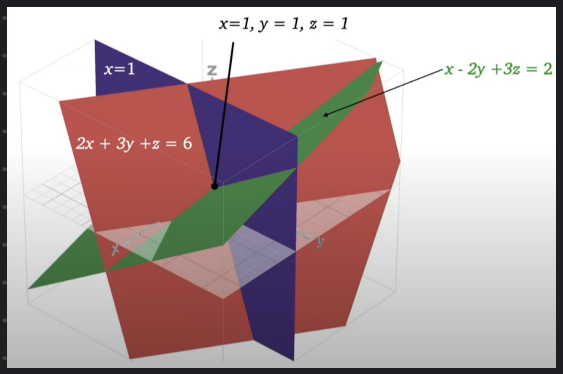
$$l_1: 2x_1 + x_2 = 3$$
$$l_2: 2x_1 + x_2 = 8$$

] only one of them can be true at a time.

Same goes with 3 unknowns:

Consider an example of 3 eqⁿs:

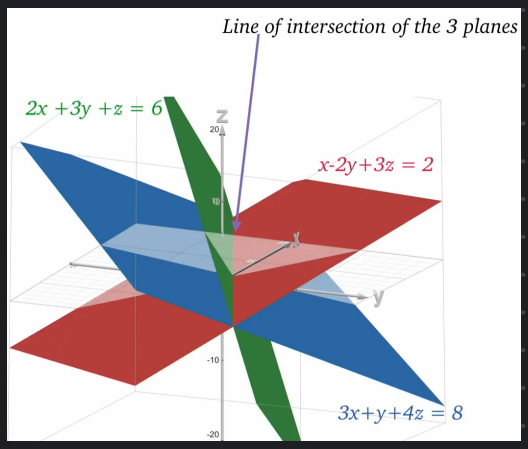
$$\begin{aligned} 2x + 3y + z &= 6 \\ x - 2y + 3z &= 2 \\ x &= 1 \end{aligned}$$



Point of intersection = (1, 1, 1)

Ex. 5.

$$\begin{aligned} 2x + 3y + z &= 6 \\ x - 2y + 3z &= 2 \\ 3x + y + 4z &= 8 \end{aligned}$$



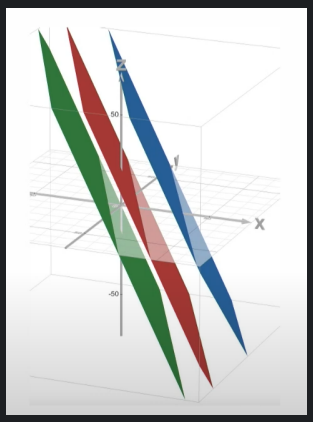
In this case, the planes intersect in a line:

$$(x, y, z) = (-\tfrac{11}{7}t + \tfrac{18}{7}, \tfrac{5t}{7} + \tfrac{2}{7}, t)$$

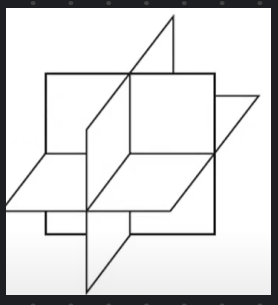
So 3 planes can intersect in a point or a line.

$$\begin{aligned} 2x + y + z &= -15 \\ 2x + y + z &= 20 \\ 2x + y + z &= 70 \end{aligned}$$

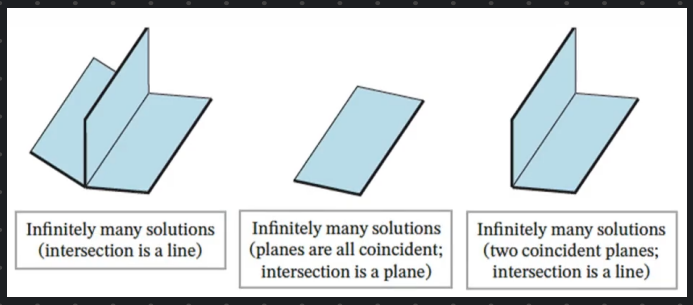
These planes are parallel.



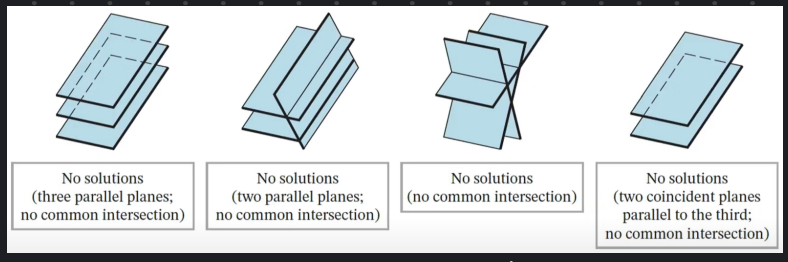
They have NO solution.



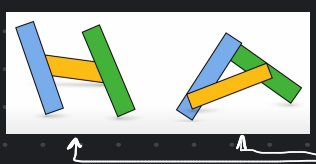
Consistent system with a unique soln.



Consistent systems with only many solns.



Inconsistent systems.



So a system of linear equations can have
- unique solution
- infinitely many solution
- no solution

Consider the system of linear equations

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned}$$

soln to this system is a sequence of n numbers
$x_1 = s_1, x_2 = s_2, \ldots, x_n = s_n$
that satisfies each and every equation in $Ax = b$.

↳ soln is an n-tuple.
complete soln set : set of all such n-tuples.

matrix forms :

$$A^{m\times n}\ x^{n\times 1} = b^{m\times 1}$$

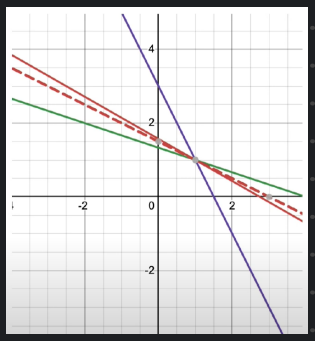
The # of eqns and # of unknowns need NOT be the same.

of eqns > # of unknowns :

$$\begin{aligned} 2x_1 + x_2 &= 3 \\ x_1 + 2x_2 &= 3 \\ x_1 + 3x_2 &= 4 \\ 4x_1 + 7x_2 &= 11 \end{aligned}$$

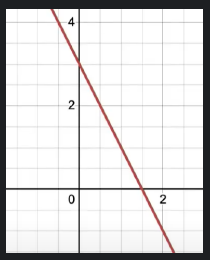
← Here # of eqns > # of unknowns.
$m > n$
Such a matrix is called a tall matrix (overdetermined system).
- more constraints than unknowns.
- soln may/may NOT exist.

Here we have a unique soln.



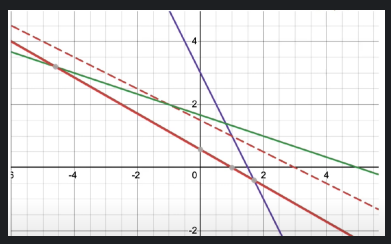
$$\begin{aligned} 2x_1 + x_2 &= 3 \\ 4x_1 + 2x_2 &= 6 \\ 3x_1 + 1.5x_2 &= 4.5 \\ 8x_1 + 4x_2 &= 12 \end{aligned}$$

← overdetermined.
All lines overlap.
∞ly many soln's.



$$\begin{aligned} 2x_1 + x_2 &= 3 \\ x_1 + 2x_2 &= 3 \\ x_1 + 3x_2 &= 5 \\ 4x_1 + 7x_2 &= 4 \end{aligned}$$

→ overdetermined system with NO solution.

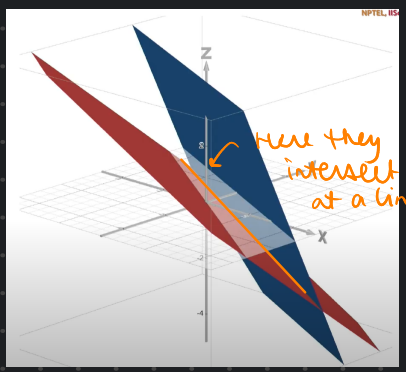


of eqns < # of unknowns : $m < n$

$$\begin{aligned} 2x_1 + 3x_2 + 4x_3 &= 0 \\ x_1 + 2x_2 + x_3 &= 3 \end{aligned}$$

← underdetermined system.

↳ eqns are 2 planes.
2 planes intersect in a line OR do NOT intersect.



Underdetermined systems do NOT have unique solns.
They have either :
- NO solns
- ∞ly many solns.

NO soln :

$$\begin{aligned} 2x_1 + 3x_2 + 4x_3 &= 0 \\ 2x_1 + 3x_2 + 4x_3 &= 3 \end{aligned}$$

Matrices

A matrix is a rectangular array of numbers in m rows and n columns.

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

Wide matrix: $m < n$

$$A = \begin{pmatrix} 1 & 0 & 2 & 5.38 \\ 2.1987 & 3 & -1 & 4 \end{pmatrix}_{2\times 4}$$

Tall matrix: $m > n$

$$B = \begin{pmatrix} 1 & 0 \\ 2 & 5.38 \\ 2.1987 & \sqrt{2} \\ 3 & -1 \\ 4 & 7 \end{pmatrix}_{5\times 2}$$

Square matrix: $m = n$

$$C = \begin{pmatrix} 1 & 0 & 2 \\ 2.1987 & \sqrt{2} & 7 \\ 3 & -17 & 0 \end{pmatrix}$$

Row matrix: $1 \times n$ or row vector

$$D = \begin{pmatrix} 1 & 0 & 2 \end{pmatrix}_{1\times 3}$$

Column matrix: or col^{m} vector

$$E = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}_{3\times 1}$$

Identity matrix: $I_{n\times n}$

$$I = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}_{I_{3\times 3}}$$

Diagonal Matrix:
- square and $d_{ij} = 0 \quad \forall \; i \neq j$

$$F = \begin{pmatrix} 4 & 0 & 0 & 0 \\ 0 & 7.2 & 0 & 0 \\ 0 & 0 & 0.123 & 0 \\ 0_{I} & 0 & 0 & \pi \end{pmatrix}$$

2 matrices are equal iff:
- size is same and
- corresponding entries are same.

$$\begin{pmatrix} 1 & 0 & 2 \end{pmatrix} \neq \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}$$

Upper triangular:
- square

AND $- \quad a_{ij} = 0 \quad \forall \; i > j$

$$U = \begin{pmatrix} 1 & 3 & 2 \\ 0 & 2 & -1 \\ 0 & 0 & \pi \end{pmatrix}$$

Lower bar: square and $a_{ij} \; \forall \; i < j$

$$L = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 2 & 0 \\ \pi & -5 & 23 \end{pmatrix}$$

Zero matrix:
- all entries are 0.
- NOT necessarily sq.

Some interpretations to solutions of systems of linear equations

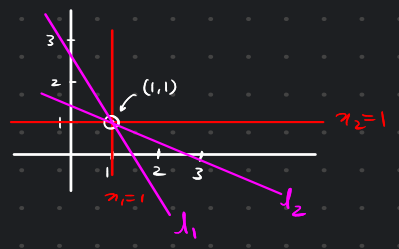
Unique soln:

$2x_1 + x_2 = 3 : l_1$
$x_1 + 2x_2 = 3 : l_2$
↳ lines that intersect at $(1,1)$

↳ coupled system: all variables contribute to all eqns.

soln: $x_1 = 1,\ x_2 = 1$ ← this itself is a system of lin. eqns.

$1x_1 + 0x_2 = 1$ ← Here any value of x_2 won't make any difference.
$0x_1 + 1x_2 = 1$ ← decoupled system. Hy no value of x_1 makes a difference.



so essentially we are trying to get from a coupled system to a decoupled system

$a_1x_1 = k_1,\ a_1 \neq 0$
$a_2x_2 = k_2,\ a_2 \neq 0$
] parallel to axes.

Both systems, the coupled one and the decoupled one, are equivalent systems.

Only many soln's:

$2x_1 + x_2 = 3$
$4x_1 + 2x_2 = 6 \rightarrow 2(2x_1 + x_2 = 3)$ Here we just get redundant information.

Redundancy: one eqn is completely specified by the other.

Inconsistent system:

$2x_1 + x_2 = 3$
$2x_1 + x_2 = 8$

The same quantities of x_1 and x_2 give different results on same operations. That cannot happen.

There's some measurement error.

Say we choose a point arbitrarily: (x_1, x_2)
we are free to choose x_1 and x_2. ↳ 2 degrees of freedom.

what if we say $2x_1 + x_2 = 3$
$\Rightarrow x_2 = 3 - 2x_1$
↳ 1 degree of freedom.

Here, when we choose a value for x_1, a value for x_2 is automatically chosen.
We can only choose 1 of (x_1, x_2) arbitrarily.
Having chosen x_1, the value of x_2 gets fixed automatically.

Say we have $2x_1 + x_2 = 3$
$x_1 + 2x_2 = 3$
↳ 0 degrees of freedom ≡ unique solution

There's only 1 pair $(x_1, x_2) = (1, 1)$ that satisfies the system.

$2x_1 + x_2 = 3$
$4x_1 + 2x_2 = 6$
] ← Here we have 2 eqns, but the same constraint is applied twice over.
1 degree of freedom.

$2x_1 + x_2 = 3$
$2x_1 + x_2 = 8$
] ← 2 constraints.
But both of them can never be met simultaneously.
No such (x_1, x_2) exist that'll satisfy the set of constraints.
NO soln's.

Matrix operations

Matrix Addition:

A and B are $m \times n$ matrices. $C = A + B$ is an $m \times n$ matrix with $c_{ij} = a_{ij} + b_{ij}$.

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 2 & -1 & 3 \end{pmatrix}, B = \begin{pmatrix} 1 & -1 & 1 & 1 \\ 2 & 3 & 4 & 5 \end{pmatrix}, C = A + B = \begin{pmatrix} 2 & 1 & 4 & 5 \\ 6 & 5 & 3 & 8 \end{pmatrix}$$

Scalar Multiplication:

Let A be an $m \times n$ matrix and k an arbitrary scalar, $C = kA$ is an $m \times n$ matrix with $c_{ij} = k \cdot a_{ij}$.

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 2 & -1 & 3 \end{pmatrix}, k = 5, kA = \begin{pmatrix} 5 & 10 & 15 & 20 \\ 20 & 10 & -5 & 15 \end{pmatrix}$$

For $k = 0$, kA is the zero matrix of the same size.

If A, B are of the same size, then $A - B = A + (-1)B$.

Properties of matrix operations:

Let $A_{m \times n}$, $B_{m \times n}$, $C_{m \times n}$ be matrices, k_1 and k_2 be scalars.

① Commutativity:
$$A + B = B + A$$

② Associativity:
$$A + (B + C) = (A + B) + C$$

③ Existence of additive identity:
$$A + O_{m \times n} = A$$

④ Existence of additive inverse element:
$$A + (-A) = (-A) + A = O_{m \times n}$$

⑤ Distributivity:
$$k_1(A + B) = k_1 A + k_1 B$$

⑥ Multiplication distributes over addn:
$$(k_1 + k_2)A = k_1 A + k_2 A$$

⑦ Associativity of scalar multiplication:
$$(k_1 k_2)A = k_1(k_2 A)$$

⑧ Identity property of scalar multiplication:
$$1(A) = A$$

Transposing a matrix:

For $A^{m \times n}$, A^T is an $n \times m$ matrix whose $(j,i)^{th}$ entry is the $(i,j)^{th}$ entry of A. $a^T_{ij} = a_{ji}$

A is symmetric if $A^T = A$. E.g. $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$

Skew symmetric: $A^T = -A$ E.g. $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ ← for a skew-symmetric matrix, diagonal elements must be 0.

Properties of Transpose:

① $(A \pm B)^T = A^T \pm B^T$

② $(A^T)^T = A$

③ $(k_1 A)^T = k_1 A^T$

④ For a square matrix, $(A + A^T)$ is always symmetric. Proof: $(A + A^T)^T = A^T + A = A + A^T$.

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \Rightarrow A^T = \begin{pmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{pmatrix}$$

$$A + A^T = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} + \begin{pmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{pmatrix} = \begin{pmatrix} 2 & 6 & 10 \\ 6 & 10 & 14 \\ 10 & 14 & 18 \end{pmatrix}$$

⑤ For a square matrix, $(A - A^T)$ is always skew symmetric. Proof: $(A - A^T)^T = A^T - A = -(A - A^T)$.

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \Rightarrow A^T = \begin{pmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{pmatrix}$$

$$A - A^T = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} - \begin{pmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{pmatrix} = \begin{pmatrix} 0 & -2 & -4 \\ 2 & 0 & -2 \\ 4 & 2 & 0 \end{pmatrix}$$

Trace of a matrix: sum of diagonal elements of A.

$\text{Trace}(A) = \text{Tr}(A) = \sum_{i=1}^{n} a_{ii}$

$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \Rightarrow Tr(A) = 1 + 5 + 9 = 15$

Properties of trace:

① For any 2 square matrices of the same size, $\text{Tr}(A+B) = \text{Tr}(A) + \text{Tr}(B)$.

② $\text{Trace}(A^T) = \text{Tr}(A)$

③ $\text{Trace}(k_1 A) = k_1(\text{Tr}(A))$ for some scalar k_1.

④ $\text{Tr}(A)$ = sum of eigen values of A.

Matrix Multiplication:

Requirement: For $A \times B$, if A has n columns, B must have n rows. $C^{m \times p} = A^{m \times n} B^{n \times p}$

Let A be a 2×3 matrix. $A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{pmatrix}$ B be a 3×2 matrix. $B = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \end{pmatrix}$

$$C = AB = \begin{pmatrix} a_{11}b_{11} + a_{12}b_{21} + a_{13}b_{31} & a_{11}b_{12} + a_{12}b_{22} + a_{13}b_{32} \\ a_{21}b_{11} + a_{22}b_{21} + a_{23}b_{31} & a_{21}b_{12} + a_{22}b_{22} + a_{23}b_{32} \end{pmatrix} \equiv \begin{pmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{pmatrix}$$

$c_{ij} = \sum_{k=1}^{3} a_{ik} b_{kj}$

In general,

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{i1} & a_{i2} & a_{i3} & \dots & a_{in} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \dots & a_{mn} \end{pmatrix} \quad B = \begin{pmatrix} b_{11} & b_{12} & \dots & b_{1j} & \dots & b_{1p} \\ b_{21} & b_{22} & \dots & b_{2j} & \dots & b_{2p} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ b_{n1} & b_{n2} & \dots & b_{nj} & \dots & b_{np} \end{pmatrix}$$

Here AB is defined but BA is not.

$$\text{then, } C = AB = \begin{pmatrix} c_{11} & c_{12} & \dots & c_{1j} & \dots & c_{1p} \\ c_{21} & c_{22} & \dots & c_{2j} & \dots & c_{2p} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ c_{m1} & c_{m2} & \dots & c_{mj} & \dots & c_{mp} \end{pmatrix}$$

$$\text{where } c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \dots + a_{in}b_{nj} = \sum_{k=1}^{n} a_{ik}b_{kj}$$

Properties of matrix multiplication:

Let A, B, C be matrices for which the foll^n operations are defined and k be a scalar. Then,

- Associative law: $A(BC) = (AB)C$
- Distributive law: $A(B+C) = AB + AC$
  $(A+B)C = (AC + BC)$
- Associative law of scalar and matrix multiplica^n: $k(AB) = kA(B) = A(kB)$
- commutativity does NOT always hold.
- If $AB = AC$, and $A \neq O_{m \times n}$, then B need NOT be equal to C. E.g.
- If $AB = BC$, then A need NOT be equal to C.
- If $AB = O^{m \times n}$, then it is NOT necessary for either A or B to be the zero matrix.

$A = \begin{pmatrix} 2 & 1 \\ 6 & 3 \end{pmatrix}, B = \begin{pmatrix} -1 & 0 \\ 5 & 2 \end{pmatrix}, C = \begin{pmatrix} 3 & 1 \\ -3 & 0 \end{pmatrix}$

$AB = \begin{pmatrix} 2 & 1 \\ 6 & 3 \end{pmatrix}\begin{pmatrix} -1 & 0 \\ 5 & 2 \end{pmatrix} = \begin{pmatrix} 3 & 2 \\ 9 & 6 \end{pmatrix}$

$AC = \begin{pmatrix} 2 & 1 \\ 6 & 3 \end{pmatrix}\begin{pmatrix} 3 & 1 \\ -3 & 0 \end{pmatrix} = \begin{pmatrix} 3 & 2 \\ 9 & 6 \end{pmatrix}$

Example: $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, B = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$, then $AB = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$

- $(AB)^T = B^T A^T$
- If A is a square matrix, then $A^0 = I$, $A^1 = A$, $A^2 = A \times A$, $A^n = A^{n-1}A$
- In general, $(AB)^2 = (AB)(AB) \neq A^2B^2$, as $A^2B^2 = AABB \neq ABAB$

Homogeneous system of equations

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n &= 0 \\ a_{21}x_1 + a_{22}x_2 + \ldots + a_{2n}x_n &= 0 \\ &\vdots \quad \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \ldots + a_{mn}x_n &= 0 \end{aligned}$$

- system of eqⁿs with RHS = 0.

$Ax = 0$

- A trivial solⁿ is all x_i's are 0 ⇒ Always consistent.
- A non-zero solⁿ is called non-trivial.
- If a homogeneous solⁿ has a unique solⁿ, it is the trivial solⁿ only.

In 2-D, 2 lines can either
- intersect at 1 point : trivial solⁿ.
- intersect at all points : non-trivial solⁿs exist.

- Homogeneous system of linear eqⁿs has
  - trivial (unique)

  OR - ∞ly many solⁿs.

NO solⁿ case does NOT exist.

Upper Δar matrices

$$A^{n\times n}\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ 0 & a_{22} & a_{23} & & a_{2n} \\ 0 & 0 & a_{33} & & \vdots \\ 0 & 0 & 0 & \ddots & \\ \vdots & & & & \\ 0 & 0 & 0 & \cdots & a_{nn} \end{bmatrix}$$



Say this is the coefficient matrix for a system of linear eq^{n}s.

Consider a 3×3 case.

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a_{22} & a_{23} \\ 0 & 0 & a_{33} \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

$a_{33}x_3 = b_3$ has only 1 variable so we can solve it.

$x_3 = \frac{b_3}{a_{33}}$ $\quad a_{33}\neq 0$

This value can be substituted in the prev. eq^n: $a_{22}x_2 + a_{23}x_3 = b_2$

Backsubstituting further would get us the desired sol^n.

A similar thing can be done with a lower Δar matrix.

For a decoupled system of linear eq^{n}s:

$$\begin{bmatrix} d_1 & & & 0 \\ & d_2 & & \\ & & \ddots & \\ 0 & & & d_n \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}$$

$d_1x_1 = c_1 \Rightarrow x_1 = \frac{c_1}{d_1}$ $\quad d\neq 0$

Ily, $x_i = \frac{c_i}{d_i}$ $\quad d_i \neq 0$

Most of the times, we can't get to such a form where we have decoupled eq^{n}s. The closest we could often go is upper Δar matrices.

Sparse Matrix:

- A matrix with only a few non-zero entries.
- E.g. consider an online purchase. There's a lot of stuff on the site. Say a matrix takes of the customer's purchase history in all items, we'd get a sparse matrix.
2. The # of places visited by a set of people.

Symmetric Matrix:

$A^T = A$

E.g. friends relation matrix over a set of n people.
→ social network graph.

Matrix multiplication
- matrix scalar multiplica^n $\quad B = kA$ means $b_{ij} = k\cdot a_{ij}$
- matrix vector multiplica^n
  - $A^{m\times n}$ $\quad B^{n\times m}$

Both AB and BA are defined here.

$A^{n\times m} \times B^{m\times n} = C^{m\times m}$



How do we pick out a particular column?

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

→ Consider the identity matrix: $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Let's make a modification $\begin{bmatrix} & & \\ & & \\ & & \end{bmatrix}$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}^{3\times3} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}^{3\times1} = \begin{bmatrix} a_{12} \\ a_{22} \\ a_{32} \end{bmatrix}^{3\times1}$$

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = a_{12}$$
$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = a_{22}$$
$$a_{31}x_1 + a_{32}x_3 + a_{33}x_3 = a_{32}$$

$$x_1 = 0, \quad x_3 = 0$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}^{3\times3} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}^{3\times1} = \begin{bmatrix} a_{12} \\ a_{22} \\ a_{32} \end{bmatrix}^{3\times1}$$